# HealthConnect Rwanda Week 7 Analytics Testing and Refinement Report

The review validates the principal source totals and strengthens the SQL treatment of missing appointment links and repeated facility-detail rows. It executed 101 original SELECT statements and 13 refinement statements in a local compatibility run. All 14 targeted refinement retests passed. Dashboard KPI and chart quality are accepted from the supplied reviewer confirmation, with static report inspection recorded separately.

## Continuity and evidence

Week 7 continues the supplied MySQL analysis, KPI script, Power BI dashboard, data-quality findings, business report and presentation. The six original files are preserved. The dashboard contains 13 pages, including the six core analytics pages, and its model diagram lists 19 tables. This report addresses analytical accuracy and interpretation rather than rebuilding the Week 6 dashboard.

Source calculations used Python and SQLite with documented MySQL function, numeric-division and header mappings. A controlled reconstruction of prepared tables supports join-grain tests. Native MySQL execution, native DAX evaluation and click-by-click Power BI interaction tests were not performed in this environment. The evidence distinguishes these limits from the reviewer acceptance of displayed KPIs and charts.

| Control | Source result | Meaning |
|---|---:|---|
| Registered patients | 20,000 | Patient register |
| Available appointments | 5,000 | Supplied appointment records |
| Patients with appointments | 4,443 | Distinct linked PatientID |
| Staff and doctors | 1,250 / 350 | One Doctor source copy |
| Reported revenue | 1,132,677,500 | Revenue Summary total |
| Collected payments | 8,389,454,556 | Payment AmountPaid total |
| Average satisfaction | 2.99 / 5 | 21,064 feedback responses |

## SQL refinements and retest results

All 41 SELECT statements in the numbered analysis and all 60 KPI SELECT statements executed in the local compatibility run. Thirty-nine numbered questions contain implemented SELECT logic; question 33 documents an unsupported service-revenue relationship. Successful execution alone does not establish that an output has the intended population or record grain.

| Component | Original result or definition | Refined result | Evidence |
|---|---|---|---|
| Laboratory volume | 2,272 linked tests | 5,000 including 2,728 Unknown | R02; RT02 |
| Department appointments | 50,000 in controlled bed join | 5,000 with unique department lookup | R01 R03; RT01 RT03 |

| Component | Original result or definition | Refined result | Evidence |
|---|---|---|---|
| Clinical-event workload | Separate grain needed | 20,000 events across four sources | R04; RT04 |
| Facility and department payments | 694,551,320 linked | 8,389,454,556 with Unknown | R05 R06; RT05 RT06 |
| Collections by month | Appointment month; linked subset | Full total by appointment cohort plus Unknown | R07; RT07 |
| Numeric rating distribution | Mean Rating by Comment | Counts of all 21,064 responses by Rating | R10; RT09 |
| Equipment distribution | 2,000 repeated IDs in controlled bed view | 200 distinct equipment IDs | R11; RT10 |
| Workforce ratio | 16 with ambiguous direction | 16 patients per staff; 62.5 staff per 1,000 patients | R12; RT11 RT12 |

## Record grain and reproducibility

The 50,000-appointment count and the 2,000-equipment count arise in the explicitly documented bed-detail reconstruction. They demonstrate why DepartmentID repeats in a bed-grain table; they are not measurements of the live imported MySQL table. R03 joins one name row per department. R11 counts distinct inventory identifiers. Department-name ambiguity was zero in the source lookup.

R04 deliberately counts clinical events rather than appointments. Its 20,000 events comprise 5,000 diagnoses, 5,000 laboratory tests, 5,000 prescriptions and 5,000 admissions. The two workload measures retain separate names and denominators. Original scripts remain in sql/original; the refined SELECT blocks are in HealthConnect_Week7_SQL_Refinements.sql.

## Financial reconciliation and attribution

| Financial control | Amount | Interpretation |
|---|---:|---|
| Total billed | 9,309,401,961 | Billing Amount |
| Collected payments | 8,389,454,556 | Payments AmountPaid |
| Outstanding balance | 919,947,405 | Billed less collected |
| Positive-balance bills | 11,887 | Bill-level aggregation |
| Appointment-linked collections | 694,551,320 | 8.28% of collections |
| Unattributed collections | 7,694,903,236 | 91.72% of collections |
| Reported monthly revenue | 1,132,677,500 | Separate 12-month summary |
| Insurance claim amount | 4,881,863,293.88 | Claims rather than cash receipts |

### The executive card and financial bars use different sources

The supplied executive Total Revenue card references Fact_Revenue_Summary [Revenue]. Its source control is 1,132,677,500. The facility and department bars reference Fact_Payments [AmountPaid], whose complete source total is 8,389,454,556. Both values can be valid when their definitions are explicit. The business report and presentation describe collected payments, so they should not be read as the expected value of the Revenue Summary card.

### Unknown attribution and rankings

The original appointment-led facility and department queries describe the linked subset. R05 and R06 begin with payments and retain the unattributed amount as Unknown, reconciling to all collections. R09 keeps Unknown in the output without ranking it as a real department. R08 ranks known facilities within the attributed subset, alongside the full-coverage control.

### Balances and payment meaning

The balance is unpaid billing under the supplied extraction, not a demonstrated operating loss. There are 39,961 differences between source billing status and the stated Paid, Partially Paid and Pending money-derived rule. Effective dates are unavailable, so preserve source statuses and derived balances separately. R13 reports payments on claim-linked bills by provider; that association does not establish the provider paid those amounts. Currency is treated as RWF from the country master because transaction currency fields are absent.

## Power BI configuration and metric scope

Static inspection found a landing page, six core analytics pages and six supplementary detail pages. Four pages carry explicit drillthrough bindings. Slicer and tooltip field definitions are present. These are configuration findings. Display quality and KPI acceptability are recorded from the reviewer statement of 2 November 2026; no native render or runtime interaction is presented as an executed test.

| Page | Purpose | Observed configuration |
|---|---|---|
| Dashboards | Landing navigation | No slicer object |
| Executive Dashboard | Core analytics | 2 slicers |
| Patient Analytics | Core analytics | 4 slicers |
| Clinical Analytics | Core analytics | 4 slicers |
| Financial Dashboard | Core analytics | 4 slicers |
| Operational Dashboard | Core analytics | 4 slicers |
| Patient Experience Dashboard | Core analytics | 4 slicers |
| Patients KPIS | Supplementary detail | No slicer object; drillthrough binding |
| Appointment KPIS | Supplementary detail | No slicer object |
| Clinical KPIS | Supplementary detail | No slicer object; drillthrough binding |
| Healthcare Staff IDS | Supplementary detail | No slicer object |

| Page | Purpose | Observed configuration |
|---|---|---|
| Operational KPIS | Supplementary detail | No slicer object; drillthrough binding |
| Financial KPIS | Supplementary detail | No slicer object; drillthrough binding |

### Labels that support interpretation

Use reported revenue for Revenue Summary totals and collected payments for AmountPaid. Patient counts based on appointment rows describe patient visits; distinct PatientID describes unique patients. Active patients in the KPI SQL means patients with at least one available appointment, without an attendance or recency definition. A repeated-admission indicator is not a clinical readmission rate without a defined event window.

The source archive contains the standard PBIX container components. The package includes an unchanged copy with a .pbix extension. It preserves all report definitions and the embedded model; no dashboard rewrite or relationship edit is claimed.

## Validated findings and business interpretation

### Patient register and recorded activity

The patient register contains 20,000 people: 10,090 Male and 9,910 Female. The available 5,000 appointments represent 4,443 distinct patients. The remaining 15,557 registered patients have no linked appointment in this extract, which does not establish that they never received care. Gender and chronic-condition segment controls reconcile to their parent populations.

| Recorded chronic condition | Female | Male | Total |
|---|---:|---:|---:|
| Asthma | 2,473 | 2,523 | 4,996 |
| Diabetes | 2,454 | 2,491 | 4,945 |
| Hypertension | 2,456 | 2,532 | 4,988 |
| None | 2,527 | 2,544 | 5,071 |

### Linkage and the documented referral count

Source checks confirm 4,708 referrals. The earlier quality PDF lists 4,709, while the business report, presentation and source data agree on 4,708. Week 7 uses the data-row count and retains the Week 6 documents unchanged. Billing has 60,000 records but only 5,000 match the available appointment parents. The Revenue Summary appointment total of 60,809 is a reported aggregate, not a maximum AppointmentID or evidence for a missing-ID range.

### Patient experience

Overall feedback averages 2.99 across 21,064 responses. Only 1,756 responses link to appointments, and that subset averages 2.93. Facility and department comparisons therefore need linked-response counts. There are 5,283 Poor comments. Without response dates or a formal complaint flag, appointment-month groupings describe a cohort proxy rather than a dated complaint series.

### Operational capacity

One Doctor source copy produces 1,250 unique staff IDs, including 350 doctors and 700 nurses. Inventory controls confirm 2,000 beds and 200 equipment IDs. All beds are Available, so occupied-status share is 0% in this snapshot.

Time-based bed, room or equipment utilization requires dated allocation or activity data. Registered-patient workforce ratios describe register coverage, not an observed staffing requirement.

## Limitations and Week 8 priorities

| Area | Remaining limitation | Recommended action |
|---|---|---|
| Appointment links | 55,000 billing and 19,308 feedback rows lack an available appointment parent | Recover an authoritative appointment export; retain Unknown meanwhile |
| Assignments | 1,187 staff and 4,744 appointments conflict with department-master facility associations | Preserve recorded IDs; resolve through an authoritative master |
| Financial timing | Transaction and status-effective dates are absent; expense data unavailable | Capture dates and settlement rules before timing or profitability claims |
| Attendance | The seven-field appointment source has no attendance/no-show target | Confirm the approved target and cohort; do not infer missed visits from missing keys |
| Capacity and feedback | Static inventory and undated feedback limit utilization and trend analysis | Capture dated occupancy, usage and feedback events |
| Execution | Prepared tables are reconstructed for controlled tests; native application runtime was not executed | Retain evidence provenance and apply the tested SQL in the actual MySQL schema |

**Resource and platform continuity**

The assignment names an approved appointment resource, data dictionary and clinic knowledge base, whereas the continued project uses the supplied Rwanda files. The guide also specifies PostgreSQL; the supplied implementation is MySQL. These differences remain explicit. No PostgreSQL conversion or backup is claimed. Clarify the accepted resource mapping with the mentor when possible.

**Actions before final integration**

- Use the refined SQL blocks with the imported table and header names, retaining the original scripts for version history.
- Keep reported revenue, collections, claims and balances separate in every explanation and export.
- Retain the current model and reviewer acceptance while preserving the distinction between static configuration and native runtime evidence.
- Prioritize appointment recovery, valid assignment mappings and dated operational events before stronger cross-component conclusions.

Cross-track collaboration has not been completed and no joint validation evidence is supplied. GitHub publication has not yet been undertaken. The prepared files support later publication and submission; those external actions are not recorded as completed.

## Coverage of the Data Analytics guide

The guide lists 18 analytical activities. Covered in scope identifies the evidence actually available: source calculations, controlled join tests, static dashboard inspection and reviewer confirmation. Attendance and cross-track activity retain their stated limitations.

| No | Activity | Coverage and evidence |
|---|---|---|
| 1 | Review Week 6 outputs | Covered. Six supplied artifacts reviewed |
| 2 | Prioritize analytical tests | Covered. T01 to T15 define scope and evidence |
| 3 | Validate KPI calculations | Covered in scope. 101 original SELECTs and source controls |
| 4 | Compare dashboard and data | Covered in scope. Static metric bindings and reviewer confirmation |
| 5 | Test filters and calculations | Covered in scope. Source segments checked; configured slicers inspected; runtime scope stated |
| 6 | Investigate inconsistent outputs | Covered. Join multiplication, linked totals and metric labels |
| 7 | Validate important findings | Covered. Patient, clinical, financial and capacity controls |
| 8 | Reassess attendance and no shows | Not assessable. No attendance outcome in the source |
| 9 | Check relevant segments | Covered in scope. Gender and chronic-condition source segments |
| 10 | Identify analytical weaknesses | Covered. Issue log I01 to I10 |
| 11 | Refine unclear visualizations | Covered in scope. Metric descriptions clarified; visual quality accepted from reviewer |
| 12 | Improve usability and interpretation | Covered in scope. Existing navigation, slicers and detail pages inventoried |
| 13 | Reassess recommendations | Covered. Recommendations reflect tested coverage and definitions |
| 14 | Identify findings for the wider solution | Covered. Key and date requirements documented |
| 15 | Share validated findings across tracks | Not completed. No actual shared validation or evidence supplied |
| 16 | Document changes | Covered. Originals preserved; refinement SQL and issue log |
| 17 | Retest important refinements | Covered in scope. 14 targeted retests passed |
| 18 | Document limitations | Covered. Report, summary and testing record |

Guide basis: Data Analytics activities on page 7 and report output on page 8; cross-track requirement on page 12; testing fields on page 15; summary and submission requirements on pages 16 and 17. The main report fulfils the report alternative. A separate Week 7 presentation is not required by that output list.

## Evidence index and handover status

| Evidence | Contents |
|---|---|
| documentation/ | Main report, project summary and detailed testing record |
| evidence/input_file_manifest.json | Six original-file hashes and package paths |
| evidence/source_verification_results.json | Source controls, assumptions, function mappings and retests |
| evidence/sql_execution_results.csv | 41 analysis, 60 KPI and 13 refined SELECT execution outcomes |
| evidence/refinement_retests.csv | 14 targeted retest outcomes |
| evidence/testing_register.csv | Objectives, scenarios, expected and actual results, actions and evidence |
| evidence/issues_and_actions.csv | Changes and remaining source limitations |
| evidence/powerbi_structure_review.json | Static page, table and field configuration inspection |
| evidence/powerbi_page_inventory.csv | All 13 pages and observed drillthrough configuration |
| evidence/reviewer_confirmation.md | Basis and scope of supplied chart/KPI acceptance |
| evidence/guide_coverage.csv | Coverage of all 18 Data Analytics activities |
| sql/ | Original SQL scripts plus the independently tested refinement blocks |
| dashboards/ | Byte-identical supplied dashboard with .pbix extension |
| week6_reference/ | Unchanged quality findings, business report and presentation |

## Submission readiness

The analytical report, summary, test record, supporting SQL, dashboard copy and evidence are prepared. Fourteen targeted retests passed. Actual cross-track validation remains not completed, and GitHub remains not yet published. Upload the required files to the designated Drive folder, check mentor access and complete the official submission process. No external upload, sharing action, commit or form submission was performed during this review.

**Source references**

The review uses the six supplied Week 6 artifacts listed in the input manifest and the original project CSVs whose hashes and row counts appear in source_verification_results. Json. The quality PDF supports the earlier issue framing; the business report and slides provide the baseline business claims; both SQL scripts and the dashboard package provide the technical definitions. The Week 7 assignment guide determines coverage and required evidence.